WORKED SOLUTION

# STEP I 1998, Question 10

Mechanics • Projectile Motion

**STEP I 1998, Question 10**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q10

## The key idea

We don't know the angle of projection, and we don't want it: the answer is in terms of $v$, $r$ and $t$ only. So look for a relation that eliminates the angle. The cleanest is to think in vectors: where a particle lands after time $t$ is its initial velocity times $t$, plus the drop $\frac{1}{2}gt^2$ caused by gravity. These three vectors form a right-angled triangle, and Pythagoras does the rest.

## Solution

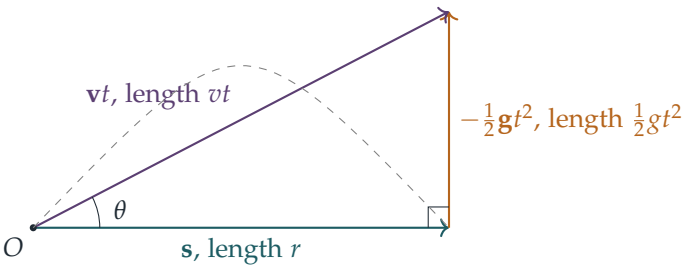
We model each particle of earth as a point particle moving freely under gravity alone (no air resistance), starting from the centre of the explosion $O$ at ground level. A particle that goes straight up lands at $O$, so we may take $r > 0$ and the projection angle $\theta$ above the horizontal with $0 < \theta < \frac{\pi}{2}$.

### The landing times

With constant acceleration $\mathbf{g}$ (vertically downwards), the displacement after time $t$ is

$$\mathbf{s} = \mathbf{v}t + \tfrac{1}{2}\mathbf{g}t^2,$$

where $\mathbf{v}$ is the initial velocity, of magnitude $v$. At the landing time, $\mathbf{s}$ is horizontal with length $r$. Rearranging, $\mathbf{v}t = \mathbf{s} - \frac{1}{2}\mathbf{g}t^2$, which is the sum of a horizontal vector of length $r$ and a vertical (upward) vector of length $\frac{1}{2}gt^2$.




These two are at right angles, so by Pythagoras

$$v^2t^2 = r^2 + \left(\tfrac{1}{2}gt^2\right)^2 = r^2 + \tfrac{1}{4}g^2t^4.$$

Multiplying by $g^2$ and writing $u = \frac{1}{2}g^2t^2$ (so that $u^2 = \frac{1}{4}g^4t^4$ and $g^2v^2t^2 = 2v^2u$), this becomes

$$u^2 - 2v^2u + g^2r^2 = 0,$$

a quadratic in $u$. Its roots are

$$\tfrac{1}{2}g^2t^2 = u = v^2 \pm \sqrt{v^4 - g^2r^2},$$

which is the required result.

**Ask yourself** Without vectors, how would we get rid of $\theta$?

Use components. Horizontally $r = vt\cos\theta$; vertically, landing at ground level means $0 = vt\sin\theta - \frac{1}{2}gt^2$, so $vt\sin\theta = \frac{1}{2}gt^2$ (as $t \neq 0$). Squaring and adding, using $\cos^2\theta + \sin^2\theta = 1$, gives exactly $v^2t^2 = r^2 + \frac{1}{4}g^2t^4$ again. The vector picture is just this calculation drawn as a triangle.

Both roots are real only when $v^4 \geq g^2r^2$, that is $r \leq v^2/g$: the familiar maximum range. When $r < v^2/g$ both roots are positive (their product is $g^2r^2 > 0$ and their sum $2v^2 > 0$), so there are two landing times, one for a steep shot and one for a shallow one. Since $t = 2v\sin\theta/g$ increases with $\theta$, the $+$ sign belongs to the steeper trajectory, $\theta > \frac{\pi}{4}$, and the $-$ sign to the flatter one.

### The greatest height

The flight is symmetric: the vertical velocity $v\sin\theta - gt'$ vanishes at $t' = \frac{1}{2}t$, halfway through the flight. From the top, the particle falls from rest (vertically) for time $\frac{1}{2}t$, so the greatest height is

$$H = \tfrac{1}{2}g\left(\tfrac{1}{2}t\right)^2 = \tfrac{1}{8}gt^2 = \frac{1}{4g}\cdot\tfrac{1}{2}g^2t^2.$$

Using the first part,

$$H = \frac{v^2 \pm \sqrt{v^4 - g^2r^2}}{4g},$$

with the $+$ sign for the steeper trajectory and the $-$ sign for the flatter one (both are possible when $r < v^2/g$; they coincide, at $H = v^2/(4g)$, when $r = v^2/g$).

## Checks

- Vertical launch: $r = 0$ gives $\frac{1}{2}g^2t^2 = 2v^2$, so $t = 2v/g$, and $H = v^2/(2g)$, which is right. The other root, $t = 0$, $H = 0$, is the grazing shot along the ground.
- The two heights add up to $v^2/(2g)$. That is as it should be: the two trajectories have complementary angles $\theta$ and $\frac{\pi}{2} - \theta$, and $\frac{v^2 \sin^2\theta}{2g} + \frac{v^2\cos^2\theta}{2g} = \frac{v^2}{2g}$.
- Numerically, with $g = 9.8$, $v = 20$ and $\theta = 1$ radian, stepping the motion gives $t \approx 3.4346$ and $H \approx 14.450$; the formulae with the $+$ sign give 3.4346 and 14.450. The complementary angle matches the $-$ sign ($t \approx 2.2053$, $H \approx 5.958$).

## Common slips

- Giving only one height. Particles with the same speed can reach the same point by two different routes, and the $\pm$ carries through to $H$.
- Writing $H$ in terms of $\theta$ and stopping. The question wants $v$, $r$ and $g$ only, so the link $H = \frac{1}{8}gt^2$ to the first part is the efficient route.

## Beyond STEP

The quadratic in $t^2$ is the time-domain version of the "envelope of trajectories": the points with $v^4 = g^2r^2$ (and, off the ground, the paraboloid $z = \frac{v^2}{2g} - \frac{gr^2}{2v^2}$) mark the boundary of the region the earth can reach, inside which every point is hit by exactly two trajectories. Envelopes of families of curves are a nice application of partial differentiation in a first-year course. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and vector kinematics at the right level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.